Explainable AI

Unit 8 · reference sheet · AI/ML in 35 topics

Axes

intrinsic (by design) vs post-hoc (after training)
global (model) vs local (one prediction)
audience decides: applicant, operator, regulator

Attribution

Saliency: $\partial f/\partial x_i$ — one pass, saturates, not complete.
Integrated gradients: $\int_0^1 \partial f(\alpha x)/\partial x_i \, d\alpha$; complete vs baseline.
LIME: sparse linear surrogate on perturbations; unstable across seeds.
SHAP: $\phi_i = \sum_{S \subseteq F \setminus i} \frac{|S|!(|F|-|S|-1)!}{|F|!}[v(S \cup i) - v(S)]$
Efficiency: $\sum_i \phi_i = f(x) - f(\text{baseline})$.

Faithfulness tests

Deletion: remove top-k attributed features ⇒ output must drop.
Insertion: add least-to-most ⇒ recovery speed.
Randomisation: shuffle weights ⇒ map must change (else it is an edge detector).
Stability: re-run explainer ⇒ same ranking?

Gotchas

- Correlation ≠ causation: attribution is about the function, not the world.
- SHAP splits credit between correlated features; neither is "the cause".
- Plausible ≠ faithful. Humans reward smooth maps.

This unit at a glance

#	Topic	You can
30	Why explainability	distinguish intrinsic vs post-hoc, local vs global; state the accountability u
31	Attribution methods	explain what gradient saliency, LIME and SHAP actually compute, and their fail
32	Evaluating explanations	test faithfulness; explain the correlation causation trap

If you see this → do that

Signal	Move
Model trusted blindly	local attribution
Regulated domain	intrinsic or SHAP + audit
Image model	saliency / IG
Tabular model	SHAP / permutation
Suspect a shortcut	deletion + randomisation

Numbers worth memorising

Shapley 2 features: $\phi_1 = 0.6, \phi_2 = 0.4$, sum $= 1.0$
Deletion top 10%: faithful ⇒ AUC $0.94 \to 0.71$
Random 3 of 20 features: accuracy ≈ 0.49 (chance)
Saliency cost: 1 backward pass; IG: 20–300